Integrate Jenkins & Ansible

- Ansible playbooks can be invoked automatically from a Jenkins job

Steps of the Integration:

- Install Ansible on a server
- Install Jenkins Ansible-Plugin
- Configure Jenkins to invoke Ansible playbook

- Both Ansible Ad-Hoc command and Ansible Playbook can be invoked from Jenkins

Exercise 38: Setup a NFS server

Implement an Ansible playbook to setup a NFS server as below steps:

IP :

NFS Server : 192.168.56.50

NFS Client : 192.168.56.51

NFS Client : 192.168.56.52

1 - Configure the Firewall (For each node)

```
yum -y install firewalld
systemctl start firewalld.service
systemctl enable firewalld.service
firewall-cmd --permanent --zone=public --add-service=ssh
firewall-cmd --permanent --zone=public --add-service=nfs
firewall-cmd --reload
```

Exercise 38: Setup a NFS server

5 - 192.168.56.51 :

vi /etc/exports

/data1 192.168.56.50(rw,sync,no_root_squash,no_subtree_check)

exportfs -arv

5 - 192.168.56.52 :

vi /etc/exports

/data1 192.168.56.50(rw,sync,no_root_squash,no_subtree_check)

exportfs -arv

7-

192.168.56.51 :

mount 192.168.56.50:/data1 /data1

192.168.56.52 :

mount 192.168.56.50:/data1 /data1

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Ansible & VMware

Ansible ships with over 50 VMware modules supporting most use cases, including:

- Managing vSphere **datacenters**, **clusters**, **hosts** and **guests**
- **VM template** and **snapshot** management
- **vSwitches**, **DNS** settings, **firewall** rules and **NAT** gateway rules

Require: python >=2.6, PyVmomi

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Vmware_datacenter

- Manage VMware vSphere Datacenters

```
- name: Create Datacenter
  vmware_datacenter :
    hostname: "{{ ansible_ssh_host }}"
    username: root
    password: 123
    datacenter_name: "MyDC"
    state: present
```

hostname

The hostname or IP address of the vSphere vCenter API server

Validate_certs

True/False. Allows connection when SSL certificates are not valid



Vmware_host

- Add/Remove an ESXi host to/from vCenter

```
- name: Add ESXi Host to vCenter
  vmware_host:
    hostname: '{{ vcenter_hostname }}'
    username: '{{ vcenter_username }}'
    password: '{{ vcenter_password }}'
    datacenter_name: datacenter_name
    cluster_name: cluster_name
    esxi_hostname: '{{ esxi_hostname }}'
    esxi_username: '{{ esxi_username }}'
    esxi_password: '{{ esxi_password }}'
    state: present
```




Vmware_guest

- **Create** new virtual machines (from templates or not).
- **Power on/power off/restart** a virtual machine.
- **Modify, rename** or **remove** a virtual machine.

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Vmware_guest(example)

```
- name: Create a VM from a template
  vmware_guest:
    hostname: 192.0.2.44
    username:
administrator@vsphere.local
    password: vmware
    validate_certs: no
    folder: /testvms
    name: testvm_2
    state: poweredon
    template: template_el7
disk:
 - size_gb: 10
     type: thin
     datastore: g73_datastore
hardware:
     memory_mb: 512
     num_cpus: 1
     scsi: paravirtual
  networks:
  - name: VM Network
     mac: aa:bb:dd:aa:00:14
  wait_for_ip_address: yes
 delegate_to: localhost
 register: deploy
```




Vmware_guest_snapshot

Create virtual machines snapshots

```
- name: Create snapshot
  vmware_guest_snapshot:
    hostname: 172.20.30.40
    username: admin@vsphere.local
    password: 123
    datacenter: DC_Name
    folder: /myfolder
    name: vm_name
    state: present
    snapshot_name: snap_name
    description: snap_desc
```

State
Present/ absent/ revert/ remove_all



Vmware_vmotion

- Move a VM to a different host

```
- name: Move a Vm to another host
  vmware_vmotion :
    hostname: 172.20.30.40
    username: admin
    password: 123
    validate_certs: False
    vm_name: my_vm
    destination_host: 172.20.40.50
```




Vmware_deploy_ovf

- deploy a VMware VM from an OVF or OVA file

```
- name: deploy an ovf
  vmware_deploy_ovf :
    hostname: 172.20.30.40
    username: admin
    password: 123
    ovf: /ymp/myvm.ovf
    datacenter: DC1
    datastore: mydatastore
    Name: myvm
    power_on: no
```




Run Ansible on Mikrotik

- /home/ansible/provision/mikrotik.yml

فایل اصلی = myProject.yml

```
---
- hosts: mikro
  connection: network_cli
  remote_user: sayahi
  gather_facts: false
  roles:
   - mikrotik
```

گروهی از mikro ها



Run Ansible on Mikrotik

- /home/ansible/provision/roles/mikrotik/tasks/main.yml

```
- name: run command on remote devices
  routeros_command:
   commands: /ip address print
  register: command_out

- debug:
   msg: '{{ command_out }}'
```




Run Ansible on Mikrotik

- /home/ansible/provision/roles/mikrotik/vars/main.yml

```
ansible_network_os: routeros
```

```
# ansible-playbook –i inventory/myhosts mikrotik.yml –k
SSH password:
        ok: [172.20.145.244] => {
          "msg": [
            [
              "Flags: X - disabled, I - invalid, D - dynamic ",
              " #   ADDRESS            NETWORK         INTERFACE                 ",
              " 0   ;;; default configuration",
              "     172.20.145.244/24  172.20.145.0    Management                ",
              " 1 X 89.165.44.238/29   89.165.44.232   Wireless                  ",
              " 2   192.168.1.1/24     192.168.1.0     Vlan194                   ",
              " 3 I 78.38.221.4/26     78.38.221.0     TCI                       ",
              " 4   192.168.3.1/24     192.168.3.0     Vlan193"
```




Run Ansible on Cisco

- /home/ansible/provision/cisco.yml

```
---
- hosts: myswitches
  connection: network_cli
  remote_user: sayahi
  roles:
   - cisco
```




Run Ansible on Cisco

- /home/ansible/provision/roles/cisco/tasks/main.yml

```
- name: run command on remote switch
  ios_command:
    commands:
      - show version
      - show interfaces
    register: command_out

- debug:
    msg: '{{ command_out }}'
```


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Run Ansible on Cisco

- /home/ansible/provision/roles/cisco/vars/main.yml

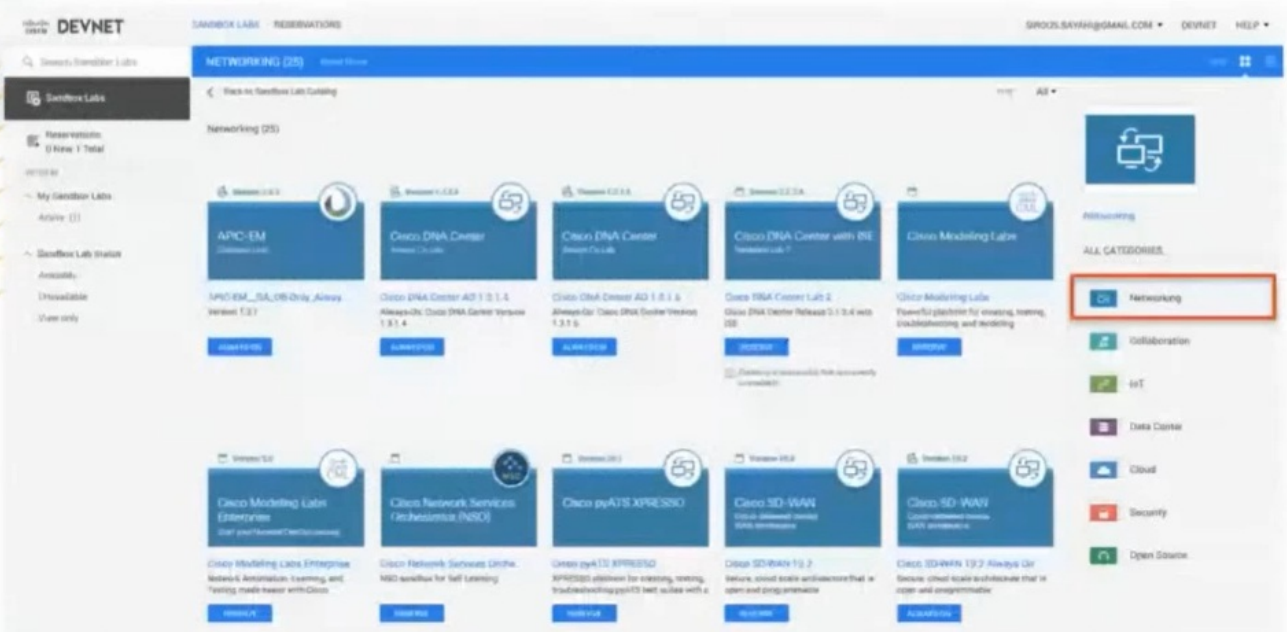
```
ansible_network_os: ios
```

```
# ansible-playbook –i inventory/myhosts cisco.yml –k
SSH password:
        ....
        "Cisco Internetwork Operating System Software ",
                        "IOS (tm) C2950 Software (C2950-I6K2L2Q4-M), Version 12.1(22)EA14,
        RELEASE SOFTWARE (fc1)",
                "Technical Support: http://www.cisco.com/techsupport",
                "Copyright (c) 1986-2010 by cisco Systems, Inc.",
                "Compiled Tue 26-Oct-10 10:35 by nburra",
                "Image text-base: 0x80010000, data-base: 0x80680000",
                "",
                "ROM: Bootstrap program is C2950 boot loader",
                "",
        ...
```




Exercise 39: Run Ansible on Cisco

- ✓ Go to https://devnetsandbox.cisco.com/
- ✓ Select the network category





Exercise 39: Run Ansible on Cisco

✓ Your lab environment also should be active now!



You can see the host address, ssh port, username and password which you can use in your Ansible playbook.



Exercise 39: Run Ansible on Cisco

✓ Run a ping-pong to check the connectivity

```
[root@Ansible provision]# ansible -m ping switches
10.10.20.100 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python"
    },
    "changed": false,
    "ping": "pong"
```




Exercise 39: Run Ansible on Cisco

✓ Run the following ad-hoc command to

[root@Ansible provision]# ansible -m ios_command -a "commands='show ip interface brief'" switches

```
"stdout_lines": [
    [
        "Interface              IP-Address      OK? Method Status                Protocol",
        "Vlan1                  unassigned      YES unset  down                  down    ",
        "GigabitEthernet0/0     10.10.20.100    YES DHCP   up                    up      ",
        "GigabitEthernet1/0/1   unassigned      YES unset  up                    up      ",
        "GigabitEthernet1/0/2   unassigned      YES unset  down                  down    ",
        "GigabitEthernet1/0/3   unassigned      YES unset  down                  down    ",
        "GigabitEthernet1/0/4   unassigned      YES unset  down                  down    ",
        "GigabitEthernet1/0/5   unassigned      YES unset  down                  down    ",
        "GigabitEthernet1/0/6   unassigned      YES unset  down                  down    ",
        "GigabitEthernet1/0/7   unassigned      YES unset  down                  down    ",
        "GigabitEthernet1/0/8   unassigned      YES unset  down                  down    ",
```


Develop your own module

✓ Uncomment the following command in ansible.cfg:

library = /usr/share/my_modules/

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✓ Develop your own module and save in the above path (e.g. test_module.py)

✓ Write your PlayBook and use the new module

Module format ...

شروع ماژول

- ✓ **Python shebang**: Every Ansible module must begin with *#!/usr/bin/python*
- ✓ **Copyright and license**: copyright line with the original copyright holder and a license declaration in ONLY one line. # Copyright: (c) 2017, [New Contributor(s)]

  # Copyright: (c) 2015, [Original Contributor(s)]

  # GNU General Public License v3.0+ (see COPYING or https://www.gnu.org/licenses/gpl-3.0.txt)
- ✓ **ANSIBLE_METADATA** block: provides information about the module for use by other tools. For new modules:

```
ANSIBLE_METADATA = {'metadata_version': '1.1',
                    'status': ['preview'],
                    'supported_by': 'community'}
```




Module format ...

- ✓ **DOCUMENTATION Block** should briefly and accurately define what each module and option does.
  - ✓ **Module**: The name of the module.
  - ✓ **Short_description**: A short description which is displayed by ansible-doc –l
  - ✓ **version_added**: The version of Ansible when the module was added.
  - ✓ **Author**: Name of the module author in the form First Last (@GitHubID).
  - ✓ **Deprecated**: Marks modules that will be removed in future releases.
  - ✓ **Options**: module options or arguments (option-name, description, required, default, choices, type, aliases, version_added, suboptions)



Module format ...

- ✓ **EXAMPLES block** show users how your module works with real-world examples in multi-line plain-text YAML format.
- ✓ **RETURN block**: documents the information the module returns for use by other modules.
  - ✓ **Description**: Detailed description of what this value represents.
  - ✓ **Returned**: When this value is returned, such as always, or on success.
  - ✓ **type**: Data type.
  - ✓ **sample**: One or more examples.

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Module format ...

```
RETURN = r'''
dest:
  description: Destination file/path
  returned: success
  type: str
  sample: /path/to/file.txt
```

✓ **Python imports:** All modules must use Python imports in the form:
from module_utils.basic import AnsibleModule



Exercise 40: use your own module

1. Place test_mod.py in /usr/share/my_modules
2. Uncomment "library = /usr/share/my_modules/" in /etc/ansible/ansible.cfg
3. Create a playbook named my_new_module
4. task/main.yml

```
- name: --- test a message ---
  test_mod:
   name: hello world

- name: --- check change ---
  test_mod:
   name: test
   new: true

- name: --- fail playbook ---
  test_mod:
   name: fail me
```

5. Run the playbook!



Use return values by other modules

```
- name: --- copy file  ---
  copy:
    src: testfile
    dest: /home
  register: copy_out
  ignore_errors: True
  tags: [mod_return]

- name: --- if the copy was copied !- - -
  debug:
    msg: 'The file was copied successfully'
  when: copy_out.failed == false
  tags: [mod_return]
```




Use return values by other modules

```
- command: /bin/false
  register: result
  ignore_errors: True

- command: /bin/something
  when: result is failed

- command: /bin/something_else
  when: result is succeeded

- command: /bin/still/something_else
  when: result is skipped
```




Exercise 41: HAProxy and nginx

- Setup and configure 2 nginx servers
  1. Install nginx
  2. Write your custom message in the index.html
  3. Start nginx service
  4. Test your nginx
- Setup and configure HAProxy server
  1. Install haproxy
  2. Configure haproxy
  3. Configure rsyslog for haproxy
  4. Start haproxy service



Exercise 41: HAProxy and nginx (solved)

```
- name: --- Add loadbalancer hostname ---
  lineinfile:
    path: /etc/hosts
    insertafter: EOF
    line: '192.168.56.102   loadbalancer'
  tags: [haproxy_hostname]
- name: --- Install nginx ---
  yum:
    name: '{{ item }}'
    state: present
  with_items:
    - epel-release
    - nginx
  tags: [Install_packages]
- name: --- update index.html ---
  copy:
        content: '<h1>Hello from
{{ ansible_hostname }}</h1>'
    dest: /usr/share/nginx/html/index.html
  tags: [update_index]
- name: --- Start nginx ---
  service:
    name: nginx
    enabled: yes
    state: started
  tags: [start_nginx]
```




Exercise 41: HAProxy and nginx (solved)

```
- name: --- Adding nginx1 and nginx2 hostnames ---
  lineinfile:
    path: /etc/hosts
    insertafter: EOF
    line: '{{ item }}'
  with_items:
    - '192.168.56.103   nginx1.loadbalancer.me   nginx1'
    - '192.168.56.104   nginx2.loadbalancer.me   nginx2'
  tags: [nginx_hostname]
- name: --- Install HAProxy ---
  yum:
    name: haproxy
    state: present
  tags: [Install_haproxy]
```




Exercise 41: HAProxy and nginx (solved)

```
- name: --- Configure haproxy (/etc/haproxy/haproxy.cfg) ---
  template:
    src: haproxy.cfg.j2
    dest: /etc/haproxy/haproxy.cfg
  tags: [config_haproxy]
- name: --- Configure rsyslog for HAProxy ---
  template:
    src: rsyslog.conf.j2
    dest: /etc/rsyslog.conf
  tags: [config_rsyslog]
- name: --- haproxy.conf fot rsyslog ---
  template:
    src: haproxy_rsyslog.conf.j2
    dest: /etc/rsyslog.d/haproxy.conf
```

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294
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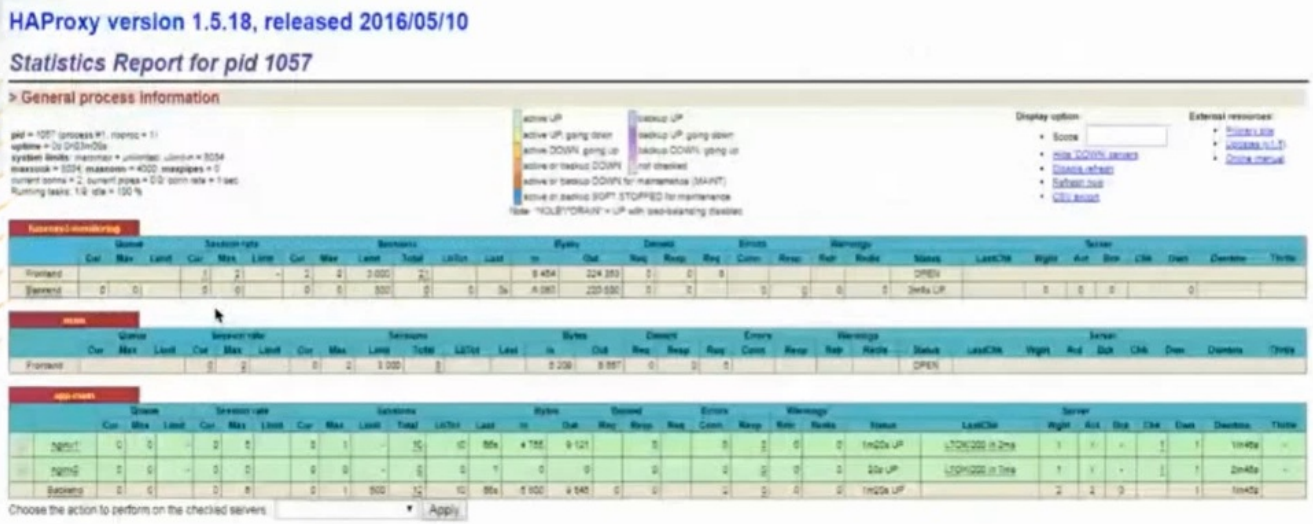
Exercise 41: HAProxy and nginx (solved)

```
- name: --- restart rsyslog service ---
  service:
    name: rsyslog
    state: restarted
  tags: [rsyslog_service]
- name: --- start haproxy service ---
  service:
    name: haproxy
    enabled: yes
    state: started
  tags: [haproxy_service]
```


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Exercise 41: HAProxy and nginx (solved)





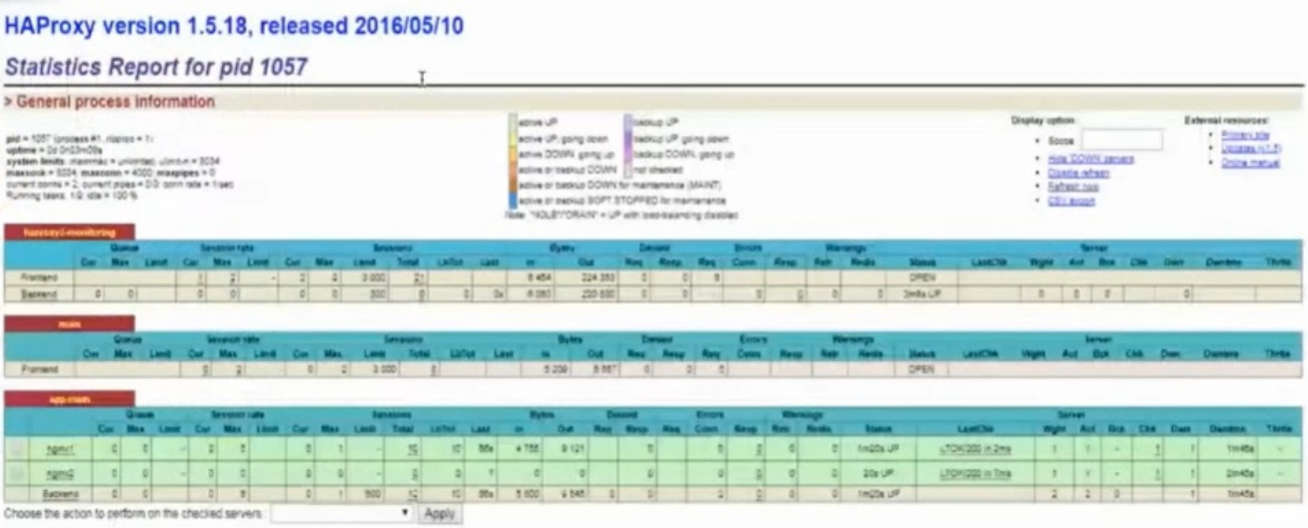
Exercise 41: HAProxy and nginx (solved)

```
- name: --- restart rsyslog service ---
  service:
    name: rsyslog
    state: restarted
  tags: [rsyslog_service]
- name: --- start haproxy service ---
  service:
    name: haproxy
    enabled: yes
    state: started
  tags: [haproxy_service]
```

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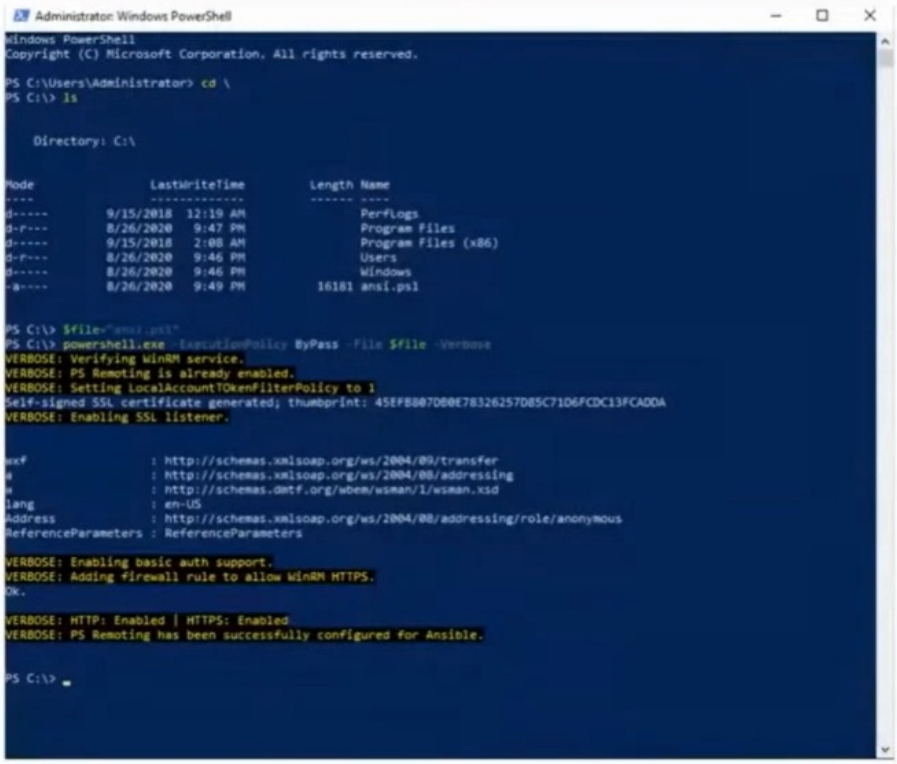
Exercise 41: HAProxy and nginx (solved)



Ansible & Windows Host

- ✓ Windows host must meet the following requirements:
- ✓ **Supported OSs**
  - Desktop include Windows 7, 8.1, and 10
  - Windows Server 2008, 2008 R2, 2012, 2012 R2, and 2016
- ✓ **PowerShell 3.0** or newer and at least **.NET 4.0**
- ✓ A **WinRM** listener should be created and activated
  - $file="ansible.ps1"
  - Powershell.exe –ExecutionPolicy ByPass –File $file -Verbose

Exercise 42: Ansible & Windows Host





Configure Ansible for windows hosts

✓ On Ansible machine (CentOS):

- Install the following packages:
- Yum install pyOpenSSL libffi-devel python-devel
- Yum install python-requests
- Yum install python-winrm



Configure Ansible for windows hosts

- ✓ Create a playbook named winprj
- ✓ Define the following variables in vars/main.yml

ansible_user: Administrator

ansible_password: qazwsx@123

ansible_port: 5986

ansible_connection: winrm

ansible_winrm_server_cert_validation: ignore

✳Check the ansible_port on the .ps1 file which you can also change it



Windows modules - win_file

- ✓ Creates, touches or removes files or directories

```
- name: ----- file module -----
  win_file:
    path: c:\dir1
    state: directory
  tags: [win_file_mod]

- name: ----- file module -----
  win_file:
    path: c:\dir1\file1.txt
    state: touch
  tags: [win_file_mod]
```




Windows modules - copy & win_feature

Win_copy Module:

```
- name: Copy a single file
  win_copy:
    src: test.txt
    dest: c:\test.txt
```

Win_feature Module:

```
- name: Install NET-Framework-Core from file
  win_feature:
    name: NET-Framework-Core
    source: D:\sources\sxs
    state: present
    restart: yes
```




Exercise 43: Install telnet client on windows

```
- name: - - - - - Install telnet-client - - - - -
  win_feature:
    name: telnet-client
    source: D:\sources\sxs
    state: present
    restart: yes
```




Windows module - win_service & win_shell

Win_service Module

```
- name: Set service startup mode to auto and ensure it is started
  win_service:
    name: W32Time
    start_mode: auto
    state: started
```

Win_shell Module

```
- win_shell: C:\somescript.ps1 >> c:\somelog.txt
```



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Windows module - win_package

```
- name: Install Remote Desktop Connection Manager locally omitting the product_id
  win_package:
    path: C:\temp\rdcman.msi
    state: present

- name: ----- Install 7-zip -----
  win_package:
    path: C:\7-Zip.exe
    product_id: 7-Zip
    arguments: /S
    state: present
```




Exercise 44: Install nodejs on windows

```
- name: ----- copy nodejs -----
  win_copy:
    src: node.msi
    dest: c:\

- name: ----- install nodejs -----
  win_package:
    path: c:\node.msi
    state: present
```

```
- name: ----- download nodejs directly --
  win_get_url:
    url: https://nodejs.org/dist/v12.18.3/
node-v12.18.3-x64.msi
    dest: c:\node.msi
```




Exercise 45: Install SQL server on windows

✓ You can run the sql server setup.exe file and follow the installation steps. In the last step you can save the configurationFile.ini without finishing the installation. So you can use this file to install sql server silently.

```
- name: --- install MSSQL ---
  win_shell: C:\sql\setup.exe /ConfigurationFile=ConfigurationFile.ini
  tags: [sql]
```




Windows module - environment variables

✓ set or remove environment variables at User, Machine or Process level.

```
- name: Set an environment variable for all users
  win_environment:
    state: present
    name: TestVariable
    value: Testvalue
    level: machine
```

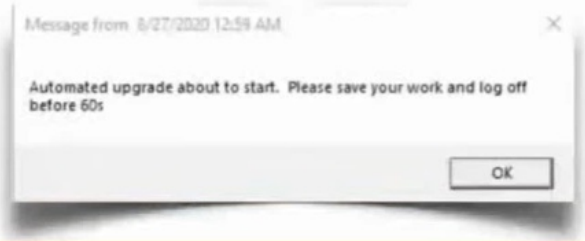
level
machine, user, process



Windows module - Sends a message to logged in users

✓ Wraps the msg.exe command in order to send messages to Windows hosts

```
- name: - - - - Warn logged in users of impending upgrade - - - -
  win_msg:
    display_seconds: 60
    msg: Automated upgrade about to start.  Please save your work
  and log off before 60s!
```



Exercise 46: Setups IIS

1. Install IIS on the windows server
2. Reboot the system if needed.
3. Copy your custom index.html into C:\\inetpub\\wwwroot\\index.html
4. Create directory c:\\inetpub\\wwwroot\\ansibletest
5. Copy index.html to C:\\inetpub\\wwwroot\\ansibletest\\index.html
6. Create a new site named Ansible Test Site on port 8080
7. Open site's port on firewall

Windows module - Sends a message to logged in users

```
- name: ----- install IIS -----
  win_feature:
    name: Web-Server
    state: present
    include_sub_features: yes
  register: win_feature
  tags: [install_iis]

- name: Reboot if installing Web-Server feature requires it
  win_reboot:
  when: win_feature.reboot_required
  Tags: [install_iis]
```


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Windows module - Sends a message to logged in users

```
- name: default-website-index
  win_copy:
    src: index.html
    dest: "C:\\inetpub\\wwwroot\\index.html"
  tags: [default_web]

- name: create new website's directory
  win_file:
    path: "c:\\inetpub\\wwwroot\\ansibletest"
    state: directory

- name: default-website-index
  win_copy:
    src: index.html
    dest: "C:\\inetpub\\wwwroot\\ansibletest\\index.html"
  tags: [default_web]
```


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Windows module - Sends a message to logged in users

```
- name: create new website
  win_iis_website:
    name: "Ansible Test Site"
    state: started
    port: 8080
    physical_path: "c:\\inetpub\\wwwroot\\ansibletest"
```

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Windows module - Sends a message to logged in users

```
- name: Open site's port on firewall
  win_firewall_rule:
    name: mysite8080
    enable: yes
    state: present
    localport: 8080
    action: Allow
    direction: In
    protocol: Tcp
    force: true
  tags: firewall
```




Windows module - Sends a message to logged in users

```
- name: create new website
  win_iis_website:
    name: "Ansible Test Site"
    state: started
    port: 8080
    physical_path: “c:\\inetpub\\wwwroot\\ansibletest"
```




Windows module - Sends a message to logged in users

```
- name: Open site's port on firewall
  win_firewall_rule:
    name: mysite8080
    enable: yes
    state: present
    localport: 8080
    action: Allow
    direction: In
    protocol: Tcp
    force: true
  tags: firewall
```




Integrate Ansible and Docker

- ✓ Setup a new VM and Install Docker
- ✓ Install **python-docker-py** on the Docker machine
- ✓ Create a playbook on Ansible server to play with Docker!



Ansible and Docker - docker_image

```
- name: - - - - - pull image - - - - -
  Docker_image:
    name: centos
    source: pull
  tags: pull_image
```

Source

build: build the image from Dockerfile
Load: load the image from a .tar file
Pull: pull the image from a registry
Local: make sure that the image is already available on the local

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